University of Bath

ME 52081 Autonomy AI/ML

Coursework – Application of Machine Learning

(Timer) Jatupoj Senarith

Word Count: 1940

1. Abstract

Human Activity Recognition (HAR) aims to identify Activities of Daily Living (ADLs) using inertial sensor data from smart devices to monitor health using assistive technologies and computational modelling. This report explores the project of developing a computational model to classify six daily activities, using data collected from initial sensors in a smartphone. The dataset used consisted of accelerometer and gyroscope recordings from multiple subjects. Three neural network machine learning models were implemented to assess the effectiveness of computational models in identifying Human Activities. The models were trained and tested using data preprocessing techniques. The results showed that all models achieved high levels of accuracy in identifying the sensor signals and classifying them into specific human activities. However, some models were more suitable due to the architectural structure of the model and its compatibility with the data. Additionally, there were misclassifications occurring in low-motion activities, leading to minimal dynamic changes in the sensor data.

2. Introduction

Human activity recognition (HAR) is a process of analysing human behaviour and movement through the usage of sensors that track movement. HAR plays a crucial role across various industries, providing endless opportunities for applications. Common examples of industrial applications include motion tracking applications for the elderly and sports analysis. With the rise in attention to health awareness, smartphones and wearable devices such as smart watches and rings have become increasingly popular, resulting in the rise in implications of HAR. The process of HAR starts with the collection of data from sensors equipped in smart devices, identifying the coordinates and movement of the device from the sensors, and the results are then processed to identify the type of human activity. As the sensors continuously collect detailed movement and changes in coordinates, machine learning methods can be implemented to determine the activity. This project focuses on applying machine learning models to the UCI Human Activity Recognition dataset [1], a dataset built from the recording of 30 subjects performing activities of daily living (ADLs) while carrying a waist-mounted smartphone (Samsung Galaxy S 11) with embedded inertial sensors. The aim is to compare and evaluate the accuracy of different classification models in order to determine which technique provides the highest accuracy and is therefore most reliable for the usage in HAR. The analytical comparison of models contributes to how effective machine learning model approaches can be for sensor data in HAR.

3. Methodology

3.1 Dataset

The data analysed in this project was from the UCI Human Activity Recognition dataset [1], which consists of 10,299 samples and 561 extracted features displaying the signals collected from the accelerometer and gyroscope sensors, representing 30 subjects performing 6 different activities, including walking, walking upstairs, walking downstairs, sitting, standing and lying down.

3.2 Data Preprocessing

The combined data was split into training (70%), validation (15%) and test (15%). A stratified sampling approach was adopted to ensure the proportions of each activity were split consistently across all subsets, to support model development and hyperparameter adjustments. The hyperparameters for each network were manually selected before the training stage including epochs, number of layers and number of neurons. *StandardScaler (Scikit learn)* was used to standardise all 561 numerical features from the dataset, which is crucial to preprocessing prior to training the model because machine learning models, including neural networks are dependent on the scale of input variables. After standardisation, the scaler was fit onto the training data and then applied to the validation test, preventing information leakage from evaluation data into the training process. Additionally, the activity labels provided in the dataset was encoded to fit the use of neural network machine learning models. In this case, the 6 human activities consisting of walking, walking upstairs, walking downstairs, sitting, standing and lying down were encoded in integers labelled 0-6. Since the UCI HAR dataset has no missing values and the data has now been set up for the applied machine learning models, the data entering the training setup has been consistently scaled and correctly formatted, no further preprocessing steps are required.

3.3 Model Architecture and Setup

To evaluate the varying accuracies of machine learning models, three neural network machine learning models were developed and trained including Artificial Neural Network (ANN), Multilayer Perceptron (MLP) and Long Short-Term Memory Network (LSTM). Training each model over twenty epochs (one complete passage of the entire dataset through the neural network during training) allows comparison of how each model has learned from the data. Although all models are representative of neural networks, each model represents how it handles the high dimensional 561 features of the dataset.

Initially, the ANN model was used to predict the six human activity classes. Despite its simplicity, the model provides effective development in non linear relationships data. As the UCI HAR dataset consists of high dimensional (561 length vector). Therefore, ANN is the most suitable initial model to assess the performance and accuracy before evaluation between other models.

The MLP model was applied to determine whether increasing the depth and additional hidden layers would generate higher accuracies within the 561 features. The deeper structure of MLP allows the model to further learn and identify complex relationships from the 6 activity classes and identify small distinctions in the data for example, walking and walking upstairs. MLP represents a complex model which allows for greater ability to distinguish patterns due to the increase in the number of neurons and layers.

In contrast to the feed forward structure of ANN and MLP models, the LSTM model assesses the patterns in the dataset using a memory based approach. The model reflects on whether the recurrent processing provides a greater accuracy in comparison to forward feed models in ANN and MLP. The data was formatted into (samples, 1, features) as LSTM traditionally requires time series sequential data.

4. Results

To evaluate the performance of each model, the validation accuracy of the prediction of each model was plotted to display which model consistently achieved the highest accuracy. Additionally, using the F-1 score which represents a performative metric used to evaluate the balance in the assessment of the results, combining recall (How many of the classes in the models were correctly identified) and precision (how many of the results were from the correct class). This section presents the evaluation of three developed neural network machine learning models ANN, MLP and LSTM.

4.1 Overview of results (UCI HAR Preprocessing Data)

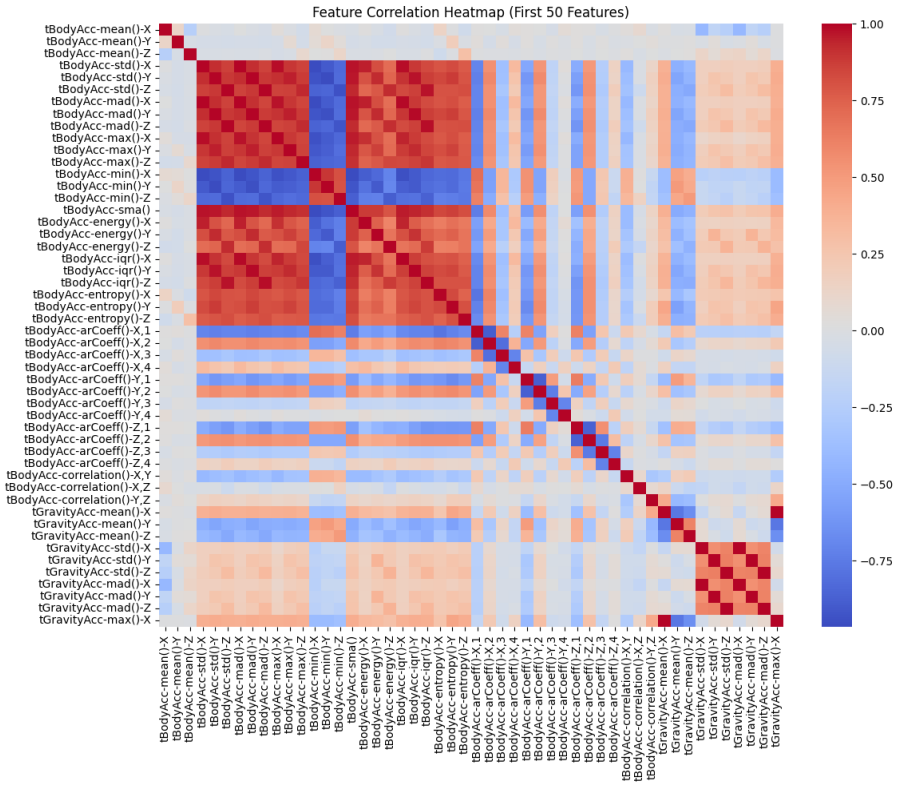


Figure 1 Correlation Heatmap

Figure 1 displays a correlation heatmap between the first fifty features of the 561 features recorded from the sensors of the smartphone. It can be seen that the data are formed in clusters throughout the heatmap, suggesting that the sensors receive similar input of motions. Therefore, neural network models are ideal for the dataset as they are capable of learning non linear relationships within high dimensional features, identifying subtle distinctions between motion signals from human activities.

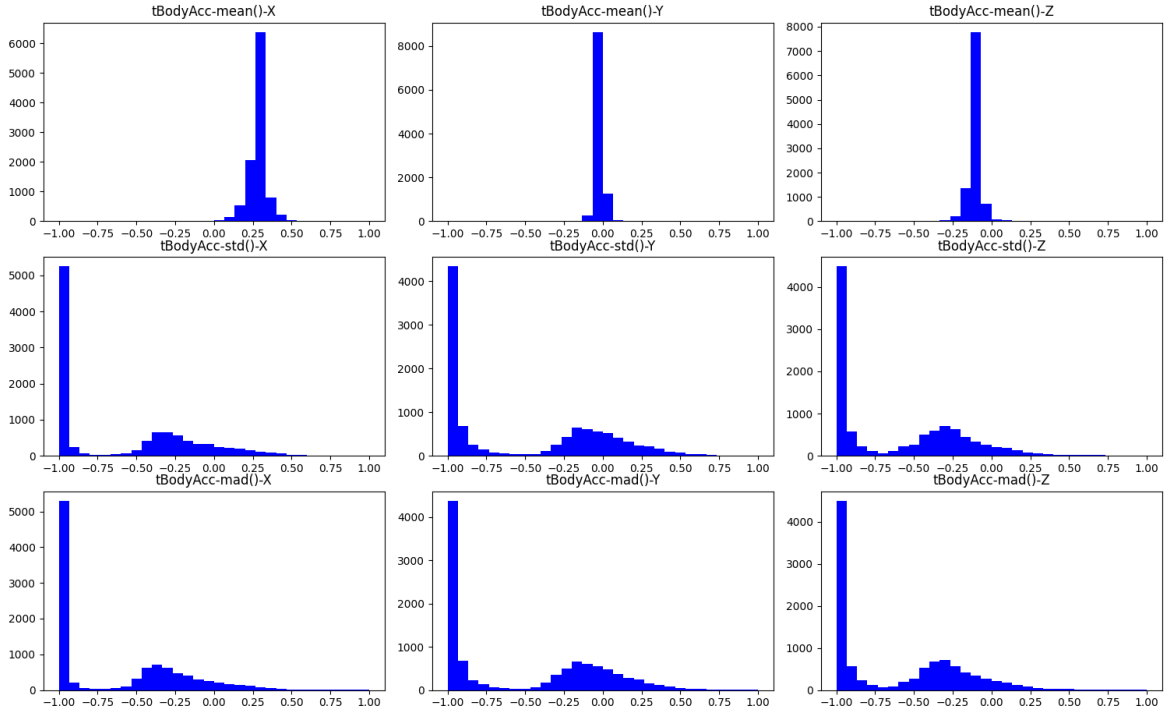


Figure 2 Histogram

Figure 2 presents a grid of histograms displaying the distribution for nine features extracted from the UCI dataset. Each histogram shows the frequency of feature values from all recorded samples after preprocessing. The distribution highlights clear differences between the X, Y and Z axis reflecting the variations in motion from the sensor. Aligning with the heatmap displayed in Figure 1, the data in Figure 2 provides the presence of complexity of the features reinforcing the compatibility of applying neural network models.

4.2 Comparative analysis between models (ANN, MLP and LSTM)

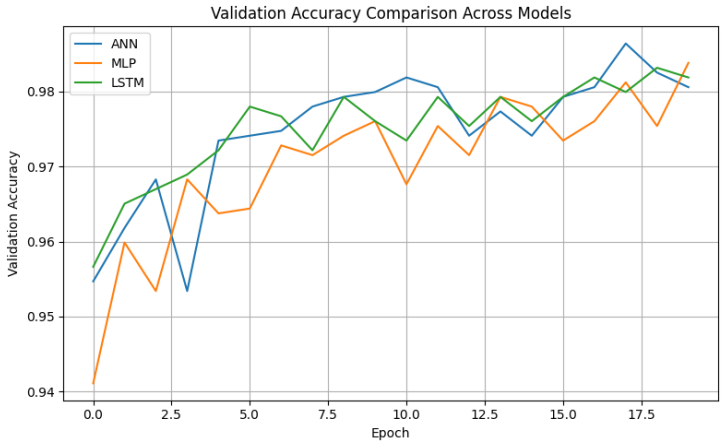


Figure 3 Validation Accuracy (All Models)

Figure 3 portrays the validation accuracy between the three models (ANN, MLP and LSTM). All three models display increasing improvements in accuracy over epochs from 94 percent to over 98, indicating effective learning from the dataset. The ANN model shows a gradual consistent increase in accuracy maintaining high performance ensuring high suitability for the complex high dimensional feature data. The MLP model reflects a similar behaviour starting from a slightly lower accuracy percentage at 94 percent in comparison to the other two models. The LSTM model displays a similar comparable validation accuracy and at certain points exceeds the levels of the two forward feeding models at the fifth epoch. Overall, the trends of all the models are marginal, therefore indicating that all models are capable of learning complex models with non linear relations in the dataset.

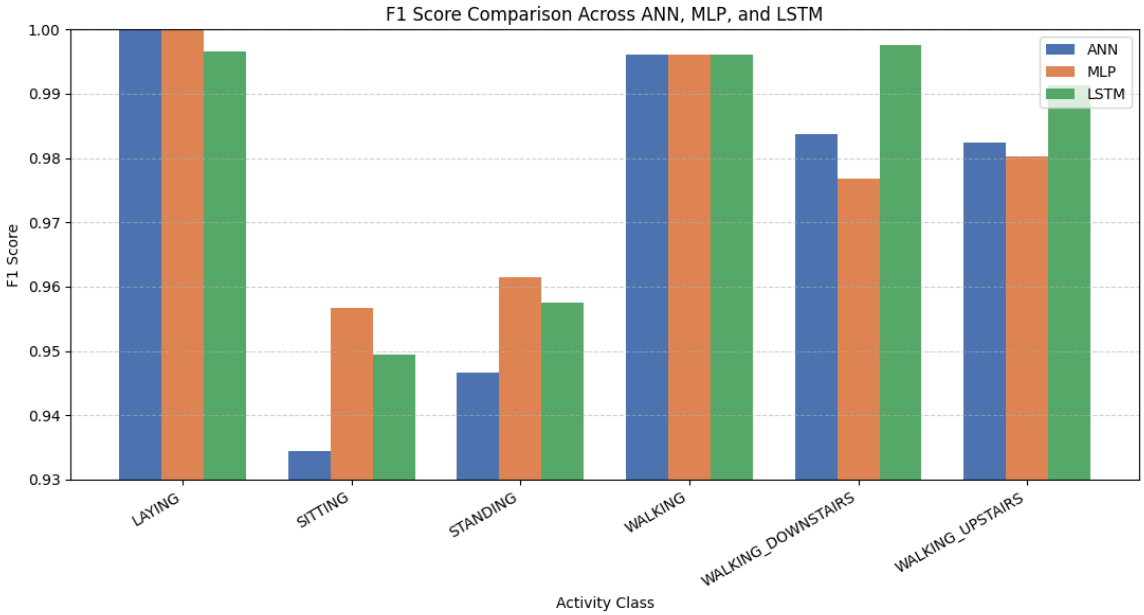


Figure 4 F-1 Activity Scores (All Models)

Figure 4 displays the F-1 score of each human activity comparison across all three models. Overall, all three models display strong classification with all scores above 0.93. Static motion activities such as lying down achieved near perfect performance scores across all models, reflecting the low variation in sensor signals. On the contrary, two activities including sitting and standing seen to display relatively lower F-1 scores across all models. This indicates the higher difficulty in differentiating lower motion activities to high movement activities. For activities involving greater movement including Walking, walking downstairs and walking upstairs, all model achieved high scores of over 0.97, most noticeably walking where all models received the same value of F-1 at just under 1.00. Overall it can be seen that all neural network models performed well on the dataset showing slight variations but displaying a trend between the motion-sensitive activities recorded by the sensor signals.

4.3 Forward-Feed models (ANN and MLP)

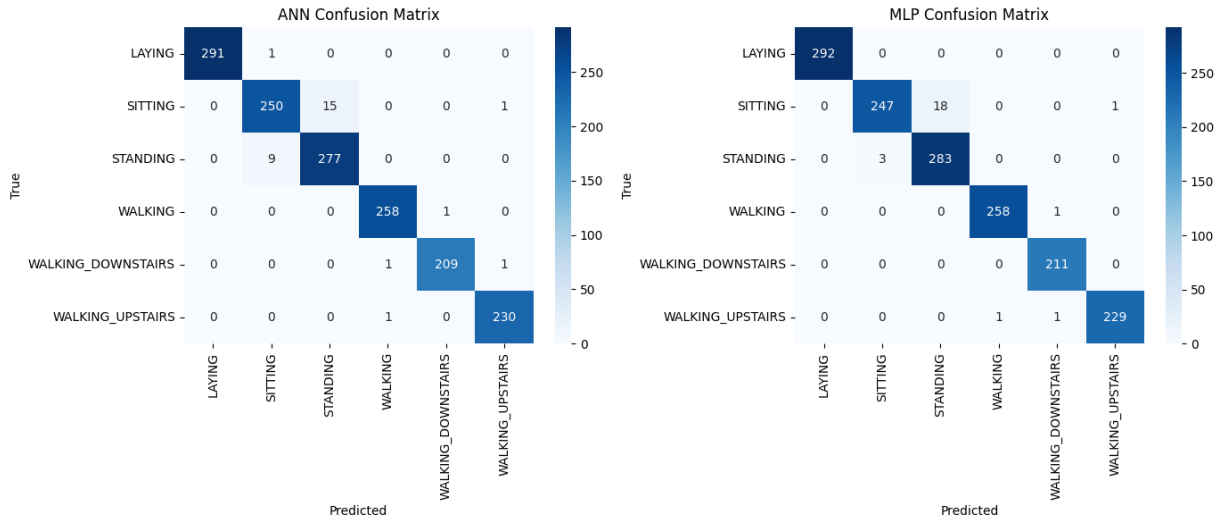


Figure 5 Confusion Matrix (ANN and MLP)

Figure 5 shows the confusion matrix from the ANN and the MLP model, displaying the performance across six human activity classes. Both models demonstrated strong capabilities for prediction, with the main diagonal indicating high accuracy. Activities, including laying and walking, displayed near perfect results by both models. For the ANN model 291 out of 292 models were correctly classified, similarly for the MLP model classified 292 correct results. However, it is notable that the majority of misclassifications appeared between sitting and standing for both models. Where for the ANN model misclassified nine sitting samples and the MLP model accordingly misclassified three sitting samples. This slight improvement could lead indications of higher performance from deeper more effective architectural models in MLP.

5. Discussion

The evaluation of three neural network models including Artificial Neural Network (ANN), Multilayer Perceptron (MLP) and Long Short-Term Memory Network (LSTM) displayed the performance of each machine learning model in predictive classification for human activity recognition (HAR). Overall, the general experimental data show that all models received high classification accuracy. Enforcing the effectiveness of neural networks model compatibility towards high dimensional based data [2]. Conversely, there are still misclassifications from the model between non-dynamic due to the lack of motion in the human activities which require improvement in processing data pipelines [3]. Despite the variation in accuracy between the models, especially between the deed-forward and LSTM, the difference in results appears to not be substantial. To summarize, although all three models provided high accuracies across all activities. However the ANN and MLP models are more suitable towards this dataset due to the fixed length data structure as LSTM requires time-series or sequential data [4]. Therefore, for ADL recognition tasks where inertial sensors have been collected in a fixed statistical data, forward-feeding neural networks are more suitable. Overall, the results demonstrate the developed computational modelling can successfully recognise the daily human activities using inertial sensor-based data from smartphones.

6. Conclusion

In conclusion, the developed recognition system successfully classified the activities of human daily living using data from inertial sensors in smartphones. Overall, the implementation in neural network models seems suitable towards developing computational analysis for HAR classification. However, there are current limitations towards the model in distinguishing less motion movements as the sensor is not able to detect them. For future work, may focus on assessing different sensors and improving sensitivity to movement to avoid misclassification in data.

7. Acknowledgements

I acknowledge that this is my own work, and I used OpenAI's ChatGPT 5.2 to assist me in Code Writing and Debugging for this coursework. (https://chatgpt.com/)

8. References

1. *Human activity recognition using smartphones*. UCI Machine Learning Repository. (Dataset)

2. LeCun Y, Bengio Y, Hinton G. *Deep learning*. *Nature*. 2015;521(7553):436–444. doi:10.1038/nature14539.

3. Anguita D, Ghio A, Oneto L, Parra X, Reyes-Ortiz JL. A public domain dataset for human activity recognition using smartphones. *Proc Eur Symp Artif Neural Netw Comput Intell Mach Learn (ESANN)*. 2013:437–442.

4. Ordóñez FJ, Roggen D. Deep convolutional and LSTM recurrent neural networks for multimodal wearable activity recognition. *Sensors (Basel)*. 2016;16(1):115. doi:10.3390/s16010115.